World Development, Vol. 17, No. 6, pp. 867–877, 1989.
Printed in Great Britain.

0305–750X/89 $3.00 + 0.00
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Property Relations and Economic Development: The Other Land Reform

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Summary. — Continued concern for development has led to the suggestion that private property rights should be created to stimulate economic development. This suggestion derives from an incomplete understanding of the property relations on the public domain lands in the arid tropics, and from a confusion of cause and effect between property and economic productivity. A model of the private-public boundary in land is developed that challenges the view that wealth would increase if land at the extensive margin were privatized. The various types of property regimes in land are defined and explained.

1. INTRODUCTION

The history of development assistance has been one of concentrated efforts to stimulate the sort of agricultural plant that exists in the donor countries — private-land based, intensive cultivation, some emphasis on mechanization, and much emphasis on the social infrastructure necessary to support that system. Where large tracts of land are under the control of a few individuals, the conventional precondition is to redistribute land to the landless so that this smallholder agriculture might be established. I will not speak to the advantages and disadvantages of this programmatic focus, nor to its record of success (or failure, for some are quite critical).

My concern here is with "the other land reform." Specifically, I am concerned with that vast domain of nonprivate land at the extensive economic margin on which millions of persons in Africa and South Asia depend for the bulk of their daily sustenance. I call it the "other land reform" to emphasize that the institutional arrangements on these lands are so little understood (and, quite often, misunderstood) that only by a careful assessment will we be able to deal properly with economic development on these lands that time, and the development community, seem to have forgotten. To consider the other land reform is to focus the mind on the institutional arrangements that may account for the current degradation of these lands, but may also — under the proper programmatic focus — hold the hope for their future.

We are not talking about trivial land areas, nor about livelihoods for the few. Recent official estimates place the standing forest cover of India at 72 million hectares (22% of the total land area), but the actual figure is said to be closer to 23 million hectares — or 7%. Whichever figure is the correct one, lands that were formerly forested are part of the public domain of interest here.[1] Degradation over the past century has reduced forest cover from an estimated 40% of what was British India (Commander, 1986). Other stories from sub-Saharan Africa tell a similar tale.

The pastoralists/nomads of sub-Saharan Africa are well-known users of lands at the extensive economic margin. But others are equally dependent upon the public domain. Recent work by Jodha in India indicates that for 21 (dry tropical) districts over seven states, between 84 and 100% of the poor households relied on public domain lands for ". . . food, fuel, fodder, and fibre items" (Jodha, 1986, p. 1172). Between 10 and 24% of the richer households in his study areas made use of the public domain for pond silt to enrich their fields, and for timber.

This vast domain in the arid tropics, on which so many individuals must depend — and usually the poorest ones at that — is the subject of a renewed interest among economists and development agencies. Drawing on the writings of biologists such as Hardin (1968) regarding the so-called tragedy of the commons, and the more ideological tracts by conservative commentators (Smith, 1981), one sees increasing belief that the

way to solve resource degradation in the arid tropics is to create private property rights in land. This interest in privatization proceeds from an incomplete understanding of the full gamut of property regimes, from a refusal to acknowledge the obvious destruction of privately-owned lands the world over (e.g., soil erosion), and from naiveté regarding who would ultimately benefit from such privatization efforts. Indeed, Jodha's research indicates that public domain lands in India have shrunk by 26–63% over the past 30 years, and that 49–86% of these privatized lands ended up under the control of the better-off segments of society. Moreover, the process of privatization was often the impetus for the destruction of the native vegetation (Jodha, 1986).

The often passionate interest in privatization of public domain lands requires careful consideration within the development community for the obvious reason that the idea of atomistic control of economic resources is so embedded in Western economic and social thought; it is so much a part of our logic and culture that to demur is to be thought blasphemous. Often, it is our most fundamental truths that warrant close scrutiny. Such is the case with development programs that emphasize privatization of the public domain in the arid tropics.

In this paper, I will discuss the notion of property rights and their role in the management of natural resources such as grazing forage, fuelwood, fodder, or timber. In Section 2, I will present a model of property rights and economic productivity that challenges the conventional wisdom that private property in land will *cause* an increase in the production from land. I will then turn to a discussion of the various types of property regimes of relevance in the arid tropics. Finally, I will close with a discussion of the relative merits of development assistance programs focused on private land versus development programs focused on the vast public domain of the arid tropics.

2. PROPERTY RIGHTS AS CAUSE OR EFFECT?

The conventional wisdom is that private property rights are a necessary condition for the generation of economic wealth; in land this means that private ownership of land is a precursor to the realization of an economic surplus. A recent interest within economics in property rights has led to the recognition of the so-called property rights school of thought, best represented by Demsetz (1967), Cheung (1970), Alchian and Demsetz (1973), and Furubotn and Pejovich (1972). This view has been reflected in recent attempts to explain the origins of agriculture (North and Thomas, 1977). The North and Thomas position is that prior to the development of the property rights paradigm there was no theory that could be used to explain the Neolithic Revolution. The property rights paradigm provided them with the foundation to suggest that it was the development of exclusive property rights over the resource base that provided a change in incentives sufficient to encourage the rise of cultivation and domestication. The view is elaborated in the following quote:

> When common property rights over resources exist, there is little incentive for the acquisition of superior technology and learning. In contrast, exclusive property rights which reward the owners provide a direct incentive to improve efficiency and productivity, or, in more fundamental terms, to acquire more knowledge and new techniques. It is this change in incentive that explains the rapid progress made by mankind in the last 10,000 years in contrast to his slow development during the era as a primitive hunter/gatherer (North and Thomas, 1977, p. 241).

This view of the origins of agriculture, both its confusion over concepts of property as well as its selective reading of history, has been challenged elsewhere (Runge and Bromley, 1979). The current task is to address the more general problem of the contemporary analogue of early hunter/gatherers — that is, those who graze animals and collect food and fiber from the vast tracts of public domain lands in the arid tropics. They are, after all, still *hunters and gatherers* — hunting for forage for their livestock, and for other food and fiber products that are collected and carried back to a home base. Our problem is to compare this ranging activity with that of sedentary agriculture of such interest to contemporary development programs.

Consider a village with a fixed population, some of whom must rely on hunting/gathering for sustenance, while the remainder can rely upon cultivated agriculture. In this model, labor is the fixed factor of production — the total hours of labor available per unit of time — while land is the variable factor. That is, the fixed labor force will range over an ever-increasing land area in search of food and fiber (or, forage for its stock). The only way in which total production of food and fiber can be increased is to range over more land, or to augment the labor force (L_2 or L_3 in Figure 1). As the necessary distances increase, notice that the production increases at a decreasing rate, reflecting the fact that the extra distances being traveled reduce the productivity of

the hunter/gatherers. The binding constraint on the production system from hunting/gathering is the spatial distribution of flora and fauna.

Now imagine that we introduce sedentary agriculture into the village economy; the question becomes one of a mix of land uses between cultivation and hunting/gathering. Continuing to hold labor constant, and letting land vary, we would obtain Figure 2; the net value of production reflects the net increment to the village over and above the necessary costs of production from the two sectors. Here, assume that the labor force (L^*) of the village is divided so that one portion works in agriculture (L^*-L_1), while the remainder (L_1) works in the hunting/gathering sector. The production function A reflects total net value of production from cultivation, while H/G_1 reflects total net value of production from hunting/gathering. The function A turns down because good quality agricultural land at the disposal of the village is in limited supply. Had the relationship been drawn in conventional terms, the productivity of workers in the two sectors would cross, indicating the proper allocation of labor between the two. This formulation simply shows the proper land allocation between the two sectors. At point B the net profitability of agriculture falls below that obtainable from hunting and gathering, even

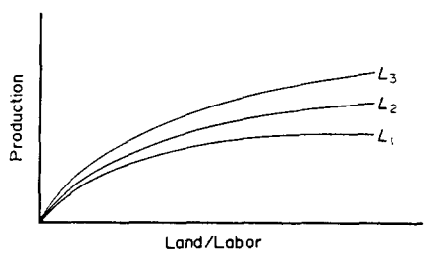

Figure 1. *Production from hunting/gathering.*

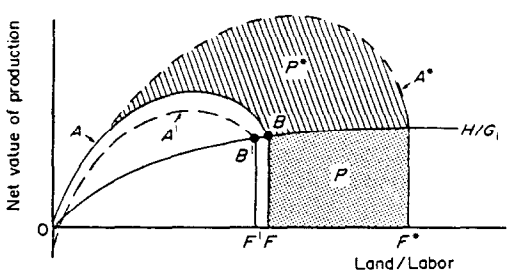

Figure 2. *The private/public frontier.*

though the net profitability of hunting/gathering is barely increasing as more land is combed for sustenance. We assume that the land in the cultivated sector is under private ownership, and that the land on which hunting/gathering occurs is in the public domain.

Those who suggest that privatization of land will bring forth new production have in mind this public domain land beyond point F — the frontier between private land the the public domain. That is, they ask us to believe that if the land out to F^* could be converted to private ownership, then A would become A^*, and the incremental net production (P^*) would exceed that now obtainable from the public domain (P). It is said to be the collective management of this public domain land between F and F^* that accounts for its low net production.

The model so far has ignored a consideration that must be reckoned with if we are to obtain a complete picture of the problem of institutional choice (type of property regime); that consideration being the costs of the particular property regime pertinent for each type of land. On the cultivated private land, we know that there must be a system of boundary surveys to demarcate the various plots of the respective owners. Additionally, there must be a process in place to record transfers of plots among new owners; this would include the measurement of the plots, the verification of a clear title to the plot, and the costs of the record keeping system in which the transaction would be recorded. Add to these official administrative costs those associated with the actions of the private owners; foremost here would be the possibility of fencing.

The administration of the public domain is not costless either. There must be meetings among the members of the village to determine the specific locations of use, to discuss rates of harvesting, and so forth. There is a reasonable case to be made that the costs of privatization (fences, measurement, title insurance, record keeping) are greater than those of the collectively-managed public domain of the village. When these hidden costs of land administration and use are incorporated into the model, we get a somewhat modified picture. Letting the costs of the public domain be a numeraire, we can depict the differential impact on the net production of the private lands by a reduction in net profit equal in magnitude to the higher infrastructure costs on the private lands. This is shown in Figure 2 with function A^1. Notice that these costs must be paid whether or not there is any production forthcoming from the private lands; A^1 is negative in the early stages. Also note that this incorporation of differential overhead costs of

private land shifts the ideal boundary point to the left (F^1).

This recognition of differential costs in two different property regimes does not really address the implicit hypothesis of those advocating extension of the private boundary into the extensive economic zone. That assumption is that privatization will *cause* production to rise by enough over and above the administrative costs that it would be wise (profitable) to extend the private/public boundary to the right. In this view, the collective management of the public domain *impedes* its natural productivity.

This is, of course, an empirical question that cannot be answered by assertions from economic theory; the mere fact that high productivity is correlated with private land does not prove that private land *causes* high productivity. Indeed, it is maintained here that the current boundary observed in the arid tropics between private land and the public domain represents the economically appropriate boundary — point F^1 in Figure 2. Beyond F^1 the net profitability from private cultivation of that land (along function A^1) would be less than that enjoyed along H/G_1 in the figure.[2]

To this point, the public domain has been treated as a single institutional entity, when in fact there are several types of property regimes that must be considered. Indeed, the confusion in the conventional literature over the "tragedy of the commons" arises from a failure to understand the concept of property, and therefore to fail to understand common property regimes.

3. ALTERNATIVE PROPERTY REGIMES

The resources of the public domain on which most villagers depend are renewable resources, capable of yielding a sustainable offtake over an indefinite time horizon. The sustainability of that use-regime is directly related to the annual harvest vis-á-vis the stock that is left behind. Humans enter these systems and take particular products that have value to them, and leave the rest. The harvesting of wood and the grazing of animals not only changes the biomass that is available for the future, but the composition of the biomass is altered as well. Indeed, the very intent of human interaction with the ecosystem is to alter the composition of that natural system to more closely conform to the products considered useful or valuable by humans.[3]

Our understanding will be enhanced if we start with the recognition that land and related resources in the rural sector are characterized by a whole complex of institutional arrangements that will vary across resources, and through seasons of the year. In some locations, or at certain times of the year, these resources may be under the control of only one individual, or one household. When this is the case, their management resembles that which is pertinent to a variety of individual (or private) resources. In other locations, or at other times of the year, however, the management of some of these resources may transcend the nominal individual or household and involve instead a number of individuals or households. Those development advisors unfamiliar with local institutional arrangements will often confuse this situation with an absence of property rights, and will then suggest that the solution is to be found in the establishment of such rights. In point of fact, as will be elaborated below, successful common-property regimes are characterized precisely by the existence of individual rights. What changes between different types of property regimes is the *scope* of the primary decision-making unit (Ciriacy-Wantrup and Bishop, 1975; Dasgupta and Heal, 1979; Netting, 1976; Rhoades and Thompson, 1975; Wade, 1987).

The essence of control over resources is that there exist socially recognized and sanctioned rules and conventions that make it clear who is the "owner" of the resource in question; call these *resource decision units*. Each decision unit will have certain interests in the management of the resource, and those interests will find expression in claims made by the decision unit (Bromley, 1989). When various claims are adjudicated and given formal protection, we say that rules and conventions are established that bestow entitlements on each decision unit. Entitlements entail a socially recognized structure of *institutional arrangements* that both constrain and liberate individuals in their behaviors with respect to other individuals; as such, institutions are at the core of group management regimes over agricultural resources. It is the institutional arrangements that comprise the binding agreement that transforms the isolation problem of a prisoners' dilemma into a cooperative game.[4] Because *property* represents a secure claim or expectation over a future stream of benefits arising from a thing or a situation (a resource, if you will), we can regard such collective management systems as *common-property regimes*. It is this failure to understand property rights that attach to resource decision units that has led to the persistent confusion between *common-property resources* and *open-access resources*, and therefore to the "inevitability of the tragedy of the commons."

A common-property regime will consist of a

well-defined group of authorized users, a well-defined resource that the group will manage and use, and a set of institutional arrangements that define each of the above, as well as the rules of use for the resource in question. In addition to the rules of use there will be rules for changing the rules of use. While it is common to refer to "common-property resources," this terminology suggests that there are certain resources that are *always* managed as common property. There are, however, no "common-property resources," just as there are no "private-property resources." There are, instead, resources that are managed as private property in one place, and as common property in another.

A resource regime derives its meaning from the structure of rights that characterize the relationship of individuals (or, as suggested earlier, primary decision units such as households or kin groups) to one another, and to the object(s) of value. The nature of these institutional arrangements defines the extant property regime over land and related resources — whether that regime be one we would call private property, or one of common property. That is, the institutional arrangements define one individual vis-á-vis others — either within the group or outside of the group. We can characterize these relations between two (or more) individuals (or groups) by stating that one party has an interest that is protected by a *right* only when all others have a *duty*. Property is a right to a benefit stream that is only as secure as the duty of all others to respect the conditions that protect that stream. When one has a *right*, one has the expectation in both the law and in practice that one's claims will be respected by those with *duty*. There are other situations in which an individual does not have a right to undertake certain actions, but instead has only *privilege*. With a right I am protected against the claim of another by their duty. With privilege I am free to do as I wish, since the other party has no rights. Put differently, an individual with privilege is free to ignore the interests or claims of those with no rights (Bromley, 1978, 1989).

The difference between a common-property regime and an open-access regime can now be made clear. In a situation of open access, I have privilege with respect to use of the resource, since no one else has the legal ability to keep me out — they have no rights. But since I have no ability to prevent them from using the resource, I have no rights, and they have privilege. It follows, therefore, that an open-access situation is one of mutual privilege and no rights. Contrast this with a common-property regime, in which there are rules defining who is in the resource management group and who is out. That is, some have a right to be in, while others have a duty to stay out. Of those recognized as being in, each has a duty to obey the rules of the group (compliance) and each has the right to expect others also to obey the rules. Here there are mutual duties and rights. It is the rights of the members limiting group size (and hence total use), along with the rights of the members proscribing the use that each will make (the stint), that together constitute property. Hence the term *common-property regime*.

Invoking the concepts of entitlements and property, it should be clear that those who confuse open-access regimes with common-property regimes, or who write of the "tragedy of the commons," have an imperfect understanding of the concept of property. If property regimes were properly understood, much mischief in the literature would never have occurred.

Before elaborating the various types of regimes for the management of agricultural resources, we should pause and mention the dynamic aspect of these legal relations. Rights and duties, as well as privilege and no rights, define individuals and groups at a particular moment in time. When an individual or a group has the legal ability to alter the status-quo structure of legal entitlements, then we say that individual (or group) has *power*, while the party who is put in a new legally binding situation has *exposure*. If a party is not able to change the legal entitlements that define it with respect to others, then that party has *no power* and the other party has *immunity* (Bromley, 1989).

When economic conditions change, or when tastes and preferences change, or when a new technique appears on the horizon, then it becomes necessary to reevaluate existing structures of entitlements (institutions) to make sure that they are not counterproductive; if these entitlements change by mutual consent we assume that both parties have been made better off, or at least one of the parties is no worse off. If, however, one party was excluded in the deliberations, or if that party was ignored or overridden, then power has been exercised.

With an understanding of right and duty, privilege and no right, it is now possible to provide a general classification of possible resource regimes, and to thereby place the common-property regime into context. I propose that, for most purposes, it is sufficient to consider four possible resource regimes: (1) state-property regimes; (2) private-property regimes; (3) common-property regimes; and (4) nonproperty regimes (or open access). Let us consider each in turn.

In a state-property regime, the control over use rests in the hands of the state. Individuals and groups may be able to make use of the resources, but only at the forbearance of the state. National forests, national parks, and military reservations are examples of state-property regimes. The nationalization of Nepal's village forests by the government in 1957 converted a common-property regime into a state-property regime.[5]

The most familiar property regime is that of private property. While most think of private property as individual property, we must remember that all corporate property is private property, yet it is administered by a group.[6] It is also essential that we be reminded of the pervasive duties that attend the private control of land and related resources; few owners are entirely free to do as they wish with such assets.

The third regime is the common-property regime. Notice that common property represents private property for the group (since all others are excluded from use and decision making), and that individuals have rights (and duties) in a common-property regime (Ciriacy-Wantrup and Bishop, 1975).

Finally we have the open-access situation in which there is no property. While the aphorism "everybody's property is nobody's property" has gained wide acclaim in the confused literature on the tragedy of the commons, this is logically inconsistent. It can only be said that everybody's access is nobody's property. The fallacy of the tragedy of the commons allegory is that by failing to understand property, and thus to see the world as dichotomous between open access (which is bad) and private property (which is claimed to be good), the commentators could leap from the presumption of destruction to the presumption of wise management, with one quick sleight of hand. The four different property regimes, and their characteristics, are summarized in Table 1. Note that it is difficult to blame the degradation of natural resources on the problem of common property since the preconditions of common-property regimes (restricted membership and use) are missing. Degradation is not the fault of the property regime, but rather of the breakdown in the incentive mechanisms necessary for the concept of property to have any meaning.

Table 1. *Four types of property regimes*

State property	Individuals have *duty* to observe use/access rules determined by controlling/managing agency. Agencies have *right* to determine use/access rules.
Private property	Individuals have *right* to undertake socially acceptable uses, and have *duty* to refrain from socially unacceptable uses. Others (called "non-owners") have *duty* to refrain from preventing socially acceptable uses, and have a *right* to expect only socially acceptable uses will occur.
Common property	The management group (the "owners") has *right* to exclude nonmembers, and nonmembers have *duty* to abide by exclusion. Individual members of the management group (the "co-owners") have both *rights* and *duties* with respect to use rates and maintenance of the thing owned.
Nonproperty	No defined group of users or "owners" and so the benefit stream is available to anyone. Individuals have both *privilege* and *no right* with respect to use rates and maintenance of the asset. The asset is an "open-access resource."

4. TOWARD IMPROVEMENT OF PROPERTY REGIMES AND RESOURCES

The recognition of three types of property regimes, and one regime that is not a property regime at all (open access), then provides a structure within which to discuss the kinds of development assistance that are most likely to succeed in the arid tropics. I will not dwell on the private land resources, since that is the subject of most development assistance programs at the current time.

Consider first the open-access regime. By definition this is a situation of mutual privilege and no right; no one user has the right (ability) to preclude use by any other party. Here the natural resource is subject to the rule of capture and belongs to no one until it is in someone's possession; whether grazing forage or fuelwood, a resource under an open-access regime belongs to the party to first exercise control over it. The investment in (or improvement of) open-access regimes must first focus on the institutional dimension. In the absence of that, and if the investment is in the form of a capital asset such as improved tree species or range revegetation, the institutional vacuum assures that use rates will eventually rise to the point of depleting the asset.

Hence with open-access resources, the necessary precondition is to convert the open-access regime to one of common property. That is, to convert mutual privilege and no-right into a situation of mutual right and duty. Once the

regime is converted to common property for the group — say all of the members of the village — then unauthorized use by outsiders will have been brought under control. An example would be if the migration patterns of herders were changed to prevent their use of village lands. The village lands would be converted from an open-access regime to one of common property for the members of the village.

With this institutional change complete, one could then begin to address the question of use rates of those within the relevant decision unit (the village). A great deal of resource degradation arises from population growth within the relevant decision unit. This use, though exceeding the ability of the renewable resource to sustain its annual yield, cannot be stopped because of the nominal right of every villager to take what he or she needs to survive. The breakdown of the common-property regime arises because of the failure of the decision-making process within the decision group. There is a failure to deal with the obvious reality that as a village grows, and therefore as the number of rights holders grows apace, the total demands on the resource will ultimately exceed its rate of regeneration. If the village believes that all of this larger population has a right to take whatever it wishes, then it becomes obvious that no villager has a right to anything other than what he or she can capture by being there first. A common-property regime for the group becomes an open-access regime for the individuals within the group.

To improve the situation requires a reduction in total offtake until the resource base can generate sufficient annual yield to meet the needs of the new (lower) harvesting, plus allow for some continued regeneration. The obvious problem is to meet the reduced needs of those deemed to be excessive claimants on the resource base until that regenerative capacity is restored. Or, if it is determined that the resource will never be able to sustain the level of demands to be placed on it, then there must be some capital investment to augment it. But capital investment in the absence of a prior institutional fix will simply assure that the new asset is squandered as the old one was.

The final resource regime upon which improvements might be concentrated is the state-property regime. The record of management here has been very disappointing (Bromley, 1986). The appearance of management by the establishment of governmental agencies, and the appearance of a coherent policy by issuance of decrees prohibiting entry to — and harvesting from — state property, has led to continued degradation of resources under the management of government agencies. When, in 1957, the government of Nepal nationalized village-level forested areas in order to bring them under more controlled management, their destruction was accelerated (Bromley and Chapagain, 1984).

If degradation of state lands is to be arrested, current practices of indifferent enforcement must be stopped (Thomson, 1977), and staffing levels and incentives must be sufficient to administer and manage that domain which the government has taken unto itself. The record to date is not encouraging.

The final issue to be addressed concerns the *economic feasibility* of investing scarce development resources and time in public domain lands vis-á-vis those lands under private control. It is obvious that, on a per acre basis, the private lands in the arid tropics are often extremely productive. Would it not make more sense to continue to invest there rather than to invest in the public domain? Or, would it not make more sense to continue to expand the irrigated area so as to convert lands out of the public domain and into the private domain?

To answer the question properly requires a number of complex considerations. But, as a brief response, several factors stand out. The conversion of land from the public to the private domain will usually require major investments in irrigation works in the face of a shadow price for foreign debt that is high and rising. Given the dismal record of production from irrigation projects in the arid tropics, the burden of expanded irrigation is further compounded (Bromley, 1982; Easter, 1977; Wade, 1975).

Second, a few considerations regarding economic feasibility seem in order. First, public domain projects must be evaluated, keeping in mind that many of the benefits will appear in the rather distant future. Under the conventional practice of discounting future benefits and costs, such projects will automatically appear to be less favorable (or less productive) than investments whose benefits appear in the nearer term. That is, most investments imply large immediate expenditures (costs) and then a stream of benefits into the future. Projects with identical net benefits over their respective lifetimes will rank differently depending upon the time stream of benefits received; those with the bulk of their benefits in the future will be discriminated against.

It is incorrect to assume that they are less productive or offer fewer benefits than projects with a different time stream of benefits; we must simply conclude that the *present value of the benefits* is different. If one adheres strictly to

conventional benefit-cost protocol, then all projects with more distant benefits will be discriminated against. It should be obvious that slavish adherence to an investment criterion that selects only those projects with the highest present-valued net benefits will result in few, if any, resource protection projects winning acceptance.

The irony of this cannot be overstated; we fail to take proper care of the livelihood and habitat of future generations because *our* valuation of the benefits is less than our valuation of projects which will take care of the present generation. The fact that we are in a position to dictate the nature of the resource endowment with which those in the future must live has not been lost on philosophers and those concerned with intergenerational ethics. It can be thought of as the tyranny of discounting. One sensible solution is to use one discount rate for project costs, and another for project benefits. The rate at which project costs would be discounted would represent the social opportunity cost of funds diverted from the private sector to the public sector. The rate at which project benefits would be discounted would be the social marginal rate of time preference. There is no reason why these two rates should be equal, and it is usually accepted that the former will exceed the latter. Marglin (1967) has suggested this approach.

Second, we must remember that benefit-cost analysis is not a dogma but is simply an evaluation tool. But being an evaluation tool does not also make it an inviolate decision rule. We must be careful to keep these two functions distinct. One can undertake a very conventional (and narrow) benefit-cost evaluation, and then employ several decision rules regarding which projects to undertake. Or, one can employ a broader and more inclusive benefit-cost evaluation and then follow a quite restrictive decision rule regarding which projects to undertake.

Third, the very concept of feasibility is a product of the economic environment within which alternative investment activities are evaluated; the context will dominate the outcome. When an evaluation of the benefits and costs of a project on private lands is undertaken, the benefit stream is largely defined by the existing infrastructure that will allow those investments to appear favorable vis-á-vis a public domain project. Marketing channels, transportation systems, agricultural extension workers, and input supply networks are usually in place and the new project is not "billed" for any of those costs; the benefit-cost analysis will usually include only the direct project costs, allowing the project a free ride on the existing infrastructure.

By way of contrast, a project on public domain lands will be less likely to benefit from such infrastructure and hence the project will necessarily reflect all of the costs related to its full realization. That this distorts the evaluation process against public domain endeavors should be obvious.

The maintained hypothesis here is that a broad-gauged evaluation of development assistance projects in the public domain would reveal that a number of endeavors are indeed economically feasible. When proper shadow pricing is undertaken on both classes of development assistance projects — that is, those on the private domain and those on the public domain — it is my view that the public domain projects will prove to be feasible locations for the concentration of institutional development and technological change. If those lands could be improved in terms of their productivity, notice that the curve H/G_1 in Figure 2 would shift up and suggest that the optimal boundary between private and public land might shift back to the left.

5. CONCLUSIONS

A fuller understanding of the institutional arrangements that define property seems necessary to a more careful assessment of the resource regimes in the developing countries. A large proportion of the population in these countries obtains a substantial fraction of their daily sustenance from lands that are *not* held in fee simple (freehold). Of particular relevance to current development concerns, those most dependent upon such lands tend to be the least advantaged members of society. The history of development assistance has been one of almost exclusive focus on private property — either in the form of more scientific agricultural practices for these lands, or in the form of land reform programs to redistribute large private estates to the landless. Few comparable efforts exist to understand the nature of economic activity on other types of lands (that is, lands under other property regimes).

The large literature on the so-called tragedy of the commons contains a fundamental asymmetry. There is ample evidence, and it is growing, of the destructive land-use practices undertaken by private (freehold) owners of many land and related natural resources. Soil erosion and deforestation are but two prominent examples.[7] In the face of this evidence, few observers challenge the ultimate social wisdom of freehold land. We are told, rather, that there must be economic incentives (bribes, tax breaks) to induce the private owner to behave in socially

acceptable ways; such bribes are often justified on the ground that the public is, somehow, "interfering" with the owner's "right" to do as he or she desires.

But when evidence of resource degradation on nonprivate land is observed, the fault is immediately said to lie with a quaint property regime that fails to assign clear ownership and, by implication, stewardship. Indeed there is almost universal agreement among development experts that the solution is to create private property for individuals, or to create state property so that the destructive users might be displaced or properly controlled by some remote government agency. We see few serious recommendations that the resource regime be studied carefully to understand the structure of rights, duties, privileges, and no rights. We see little recognition that resource regimes in the developing countries might be revitalized by efforts to understand their systemic problems, as opposed to expropriation of the resource by the state. In short, there seems little recognition of the rich array of institutional alternatives whereby land and related resources might be managed.

This continuum of choice and control — ranging between complete individualization and total collectivization — is the subject of great intellectual and popular debate. Indeed, the debate goes back to the Greek philosophers (Schlatter, 1951). The essence of contemporary economics is to celebrate the individual as the proper decision maker, and from that it follows that the individual must have full control over the necessary productive inputs (of which land and related natural resources are central). In conventional economic theory we start with the assumption that all valuable resources are individually owned, fully mobile, and exchangeable in small increments in well-functioning markets. We then conclude that these conditions will assure an efficiently operating system. Notice that the paradigm demands the very premises that allow us to reach such happy conclusions regarding efficiency. That is, we say that *if* the right conditions exist — divisibility, mobility, full ownership of all valuable resources, full information, and well-functioning markets — then the allocation of all factors of production will be efficient. As a definition this is fine; as policy prescription in a world where not all valuable resorces are fully divisible and capable of individual ownership, where all resources are not fully mobile, where information is imperfect, where many markets — but especially contingent-claims markets — are not present, and where the future is unrepresented, then our standard advice is suspect. These conditions that render facile privatization recommendations so misguided are precisely the conditions that characterize natural resource use in the developing countries.

The mischief to arise from such policy pronouncements can be attributed to the fact that those giving advice do not always understand the critical distinction between open-access resources (*res nullius*) and common-property resources (*res communis*). Open access is a free-for-all, common property represents a well-defined set of institutional arrangements concerning who may make use of a resource, who may *not* make use of a resource, and the rules for how the accepted users shall conduct themselves. By failing to understand this essential difference, many then commit an equally serious mistake. That is, institutional arrangements over natural resources (property arrangements) are thought to be at two polar extremes; one either has individual (or private) property, or one has a free-for-all.

By failing to understand that property arrangements are not bimodal but in fact are to be found along a continuum, observers are led to embrace one pole on the evidence of imminent disaster at the other. Moreover, such reasoning ignores centuries of wise use of commonly-controlled resources in the Swiss Alps (Netting, 1976), in the Andes (Rhoades and Thompson, 1975), and in other locations where exogenous technology and market processes have not disrupted the human-ecosystem interaction. Third, such reasoning invites incredulity in the developing world when various cultures are told that their salvation lies in the extreme individualization of all resource decisions. Finally, and possibly most damaging, the record of resource use in countries where individualization is present and revered would lead the neutral observer to ponder the ultimate wisdom of unconstrained atomization of control.

Development assistance programs that begin to address *the other land reform* would open up a range of meaningful endeavors that would seem to hold considerable promise for the natural resources of the arid tropics, and for the millions of poor people who must now depend upon these lands.

NOTES

1. I will use the term "public domain" here as shorthand for all of those lands held in *other than* freehold (fee simple). I will classify types of ownership regimes later in the paper. The conventional view is that

approximately one-half of India is uncultivated — and hence public domain land (Bentley, 1984).

2. The presence of irrigation is one obvious way in which the optimal boundary can be adjusted in the arid tropics. In the absence of irrigation however, the presumption must be that the village has located the efficient boundary between private and public land.

3. See the work on coevolutionary development by Norgaard (1981).

4. The prisoners' dilemma is a situation in which each of the participants has a consistent tendency to defect from an agreement and to seek to better her or his own situation at the ultimate expense of the others in the group. This tendency to defect means that the aggregate of all participants is less well off than if the individuals had stuck together. See Friedman (1986); Guttman (1978); Schotter (1981); or Shubik (1982).

5. Although, as will be suggested below, in the absence of consistent administration and enforcement the forests became open-access regimes.

6. The important difference between corporate property and common property, even though both are "private property" for the group, is that in common-property regimes the owners are actively engaged in use and control of the asset. Corporate property usually has intervening parties (management) that control use and decision making.

7. It is common in the United States for state laws to require the reforestation of private lands following the harvest of timber; laws which would presumably not be necessary if private owners had not been prone to clear-cut lands and then leave them to erode.

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